Explainable-AIOps: Normal vs Slow Traffic Evaluation

Model: normal_extended_v1 — LSTM Autoencoder

Purpose: Compare reconstruction-error-based anomaly flags for the saved held-out normal test dataset and the isolated slow-traffic dataset.

1. Experiment Configuration

Parameter	Value
Model experiment	normal_extended_v1
Window size	10
Features per time step	12
Anomaly threshold	0.025898804888
Threshold method	validation_reconstruction_error_percentile
Model retrained for this report	No

2. Dataset Comparison

Metric	Normal test	Slow traffic
Windows evaluated	81	31
Windows flagged	1	31
Windows not flagged	80	0
Flagged percentage	1.23%	100.00%
Mean reconstruction error	0.01152777	173.37988548
Median reconstruction error	0.00816641	174.06690000
Minimum reconstruction error	0.00216976	171.53279000
Maximum reconstruction error	0.02621923	175.44173000

3. Interpretation

The saved normal test results contain 81 windows, of which 1 exceeded the threshold (1.23%). The isolated slow-traffic results contain 31 windows, of which 31 exceeded the threshold (100.00%).

These results show how this model behaved on these particular datasets. They do not establish general production accuracy or prove that every flagged window corresponds to a real incident.

4. Limitations

• The normal and slow-traffic datasets are different samples and have different numbers of windows.

• The slow-traffic dataset contains only 40 observations and produces 31 overlapping windows.

• Overlapping windows share observations and are not statistically independent.

• The threshold was selected from normal validation reconstruction errors; it was not adjusted for this comparison.

• A normal-only test set estimates false alarms on that sample but does not measure anomaly recall. Independent, labeled datasets are needed to assess broader performance.

• Reconstruction error is an anomaly indicator, not an explanation of the operational root cause.

Source files: test_reconstruction_errors.csv, slow_traffic_results.csv, anomaly_threshold.json, and experiment_summary.json. This report was generated from existing saved results; it does not retrain or modify the model.